A uniform Kolmogorov bound for Jack measures

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Abstract

For the normalized transposition statistic under Jack measure on partitions of n, we prove a Kolmogorov bound $10\max\{n^{-1/2}, \sqrt{\alpha}/n\}$ for every $n \geq 3$ and $\alpha \geq 1$. This removes the logarithmic loss in the known bound, including the transition $\alpha \asymp n$, and proves the uniform estimate conjectured by Fulman. The proof retains cancellation in the conditional third moment of a Jack growth exchangeable pair. Its expansion in three orthogonal Jack characters gives the required parameter dependence.

1 Introduction

Jack measures deform Plancherel measure on integer partitions and provide discrete analogues of Gaussian beta ensembles [2]. For a partition $\lambda \vdash n$, put

$$c_\lambda = \prod_{u\in\lambda} (\alpha a(u) + \ell(u) + 1), \qquad c'_\lambda = \prod_{u\in\lambda} (\alpha a(u) + \ell(u) + \alpha), \qquad \pi_{n,\alpha}(\lambda) = \frac{\alpha^n n!}{c_\lambda c'_\lambda}.$$

Here $a(u)$ and $\ell(u)$ count the boxes to the right of and below u. The parameter satisfies $\alpha > 0$; at $\alpha = 1$ the law is Plancherel measure. Write λ_i and λ'_i for row and column lengths. The statistic of interest is

$$W_{n,\alpha} = \frac{\sum_i \left[\alpha\binom{\lambda_i}{2} - \binom{\lambda'_i}{2}\right]}{\sqrt{\alpha\binom{n}{2}}}, \qquad \lambda \sim \pi_{n,\alpha}. \tag{1}$$

It has mean zero and variance one [2, Lemmas 3.1–3.2]. At $\alpha = 1$, it is the normalized irreducible character ratio of a transposition. Its normal approximation quantifies fluctuations of this character statistic as the partition size and the deformation parameter vary.

Let Z be standard normal, with distribution function Φ, and write

$$d_{\mathrm{K}}(V, Z) = \sup_{x\in\mathbb{R}} |\mathbb{P}(V \leq x) - \Phi(x)|.$$

Fulman conjectured a bound by a universal constant times $\max\{n^{-1/2}, \sqrt{\alpha}/n\}$ [2, p. 4 of the author version]. Chen, Raič and Thành proved $8.2\max\{n^{-1/2}, \sqrt{\alpha}\log n/n\}$ [1, Proposition 4.1]. Their estimate has the desired order for $1 \leq \alpha \leq n/\log^2 n$; they also obtained the desired order for $\alpha \geq n^{1+\varepsilon}$, with a constant depending on the fixed $\varepsilon > 0$. Thành and Tu's subsequent non-uniform estimate applies on the restricted range $\log^2 n/n \leq \alpha \leq n/\log^2 n$ [6, Theorem 8]. The uniform Kolmogorov estimate across the intermediate range is the remaining question. The following bound covers the whole parameter range.

Theorem 1.1. *For every integer $n \geq 3$ and every real $\alpha \geq 1$,*

$$d_{\mathrm{K}}(W_{n,\alpha}, Z) \leq 10 \max\left\{ n^{-1/2}, \frac{\sqrt{\alpha}}{n} \right\}. \tag{2}$$

The constant is independent of both parameters; no restriction is needed when α is comparable to n. We do not optimize its numerical value. The growth coupling of Kerov [3] supplies an exchangeable pair whose increment need not have a small deterministic bound. We use the normal-approximation inequality of Shao and Zhang [5], which depends on the signed conditional third moment rather than the absolute third moment of the increment. We compute this conditional moment as a linear combination of three orthogonal Jack characters. Orthogonality then gives its exact second moment and the uniform estimate.

2 Jack growth and the cubic increment

2.1 Corner probabilities and content moments

Fix $\alpha \geq 1$ and write $d = \alpha - 1$. The content of a box (i, j), with indices starting at one, is $c(i, j) = \alpha(j - 1) - (i - 1)$. For any partition ρ, set

$$s_k(\rho) = \sum_{u \in \rho} c(u)^k.$$

The contents sum to the numerator of (1), by summing $j - 1$ along rows and $i - 1$ along columns. An outer corner has coordinate $x = c(u)$ for an addable box u. An inner corner has coordinate $y = c(u) + d$ for a removable box u; the shift arises because the inner coordinate uses the south-east corner of the box. Define

$$G_\rho(z) = \frac{\prod_{y \text{ inner}}(z - y)}{\prod_{x \text{ outer}}(z - x)}. \tag{3}$$

Kerov's growth process adds a box to ρ with content X, with probabilities equal to the residues of G_ρ. If λ has n boxes, its reverse process deletes a box of content Y, and

$$G_\rho(z) = \mathbb{E}\left(\frac{1}{z - X} \,\middle|\, \rho \right), \qquad \frac{1}{G_\lambda(z)} = z - \alpha n \mathbb{E}\left(\frac{1}{z - (Y + d)} \,\middle|\, \lambda \right). \tag{4}$$

These are the transition and cotransition formulas in [3, Section 2, Theorem 6.7 and Lemma 7.2]. Both formulas use exactly the content and shifted inner-corner coordinates defined above.

Both kernels have the Jack laws as coherent marginals. To check the normalization and reversal explicitly, let $\kappa(\nu, \lambda)$ be the positive one-box Jack Pieri multiplicity in Kerov's notation, and put $D(\rho) = |\rho|! \alpha^{|\rho|} / c'_\rho$ and $\varphi(\rho) = 1/c_\rho$. The hook and branching formulas give

$$D(\lambda) = \sum_{\nu \nearrow \lambda} \kappa(\nu, \lambda) D(\nu), \qquad p(\nu, \lambda) = \kappa(\nu, \lambda) \frac{\varphi(\lambda)}{\varphi(\nu)}, \qquad q(\lambda, \nu) = \kappa(\nu, \lambda) \frac{D(\nu)}{D(\lambda)}$$

[3, Sections 6–7]. Since $\pi(\rho) = D(\rho)\varphi(\rho)$,

$$\pi_{n-1,\alpha}(\nu) p(\nu, \lambda) = \pi_{n,\alpha}(\lambda) q(\lambda, \nu). \tag{5}$$

Starting from the empty diagram and summing proves the marginal assertion and normalization at every level.

The two transforms give the low moments needed below.

Lemma 2.1. *If ρ has m boxes, upward growth satisfies*

$$\mathbb{E}(X \mid \rho) = 0, \quad \mathbb{E}(X^2 \mid \rho) = \alpha m, \quad \mathbb{E}(X^3 \mid \rho) = 2\alpha s_1(\rho) + \alpha d m. \tag{6}$$

If λ has n boxes and $N = \binom{n}{2}$, reverse growth satisfies, with $s_k = s_k(\lambda)$,

$$\mathbb{E}(Y \mid \lambda) = \frac{2s_1}{n}, \tag{7}$$

$$\mathbb{E}(Y^2 \mid \lambda) = \frac{3s_2 - ds_1 - \alpha N}{n}, \tag{8}$$

$$\mathbb{E}(Y^3 \mid \lambda) = \frac{4s_3 - 3ds_2 + d^2 s_1 - 2\alpha(n-1)s_1 + \alpha d N}{n}. \tag{9}$$

Proof. Row-wise telescoping in (3) gives

$$G_\rho(z) = \frac{1}{z} \prod_{u \in \rho} \frac{(z - c(u))(z - c(u) - d)}{(z - c(u) - \alpha)(z - c(u) + 1)}. \tag{10}$$

Indeed, row i of length r contributes $(z+i-1)(z+i-\alpha r)/[(z+i)(z+i-1-\alpha r)]$. The first factors telescope over rows; the others cancel across equal successive row lengths, leaving the genuine corners in (3).

In the following display $s_k = s_k(\rho)$. Write L_j for the coefficient of z^{-j} in the formal Laurent series $\log(zG_\rho(z))$. Its z^{-1} coefficient vanishes, and a box of content c contributes $[(c+\alpha)^j + (c-1)^j - c^j - (c+d)^j]/j$. Hence

$$L_2 = \alpha m, \qquad L_3 = 2\alpha s_1 + \alpha d m,$$
$$L_4 = 3\alpha s_2 + 3\alpha d s_1 + (\alpha d^2 + \alpha^2/2)m,$$
$$L_5 = 4\alpha s_3 + 6\alpha d s_2 + (4\alpha d^2 + 2\alpha^2)s_1 + (\alpha d^3 + \alpha^2 d)m.$$

The first transform in (4) proves (6). For the second transform take $\rho = \lambda$ and $m = n$, and expand

$$\frac{1}{G} = z - L_2 z^{-1} - L_3 z^{-2} - (L_4 - L_2^2/2)z^{-3} - (L_5 - L_2 L_3)z^{-4} + O(z^{-5}).$$

It follows that the first three moments of $Y + d$ are, respectively, $L_3/(\alpha n)$, $(L_4 - L_2^2/2)/(\alpha n)$ and $(L_5 - L_2 L_3)/(\alpha n)$. Expanding $Y = (Y + d) - d$ gives (7)–(9). All expansions here are formal coefficients of finite rational functions. □

2.2 Three orthogonal Jack characters

We now assume $n \geq 5$, and keep $N = \binom{n}{2}$. To estimate the cubic expression under Jack measure, use the coefficients of power sums. Here $p_r = \sum_i x_i^r$ in formal variables and $p_\mu = \prod_{j=1}^{\ell(\mu)} p_{\mu_j}$, with $\ell(\mu)$ the number of parts of μ. Expand the integral Jack polynomial as $J_\lambda^{(\alpha)} = \sum_{\mu \vdash n} \theta_\mu^\lambda(\alpha) p_\mu$. Write $\theta_k = \theta^\lambda_{(k,1^{n-k})}(\alpha)$ and

$$\delta = \frac{\alpha - 1}{\sqrt{\alpha}}, \quad b = \delta^2, \quad P_k = \frac{s_k(\lambda)}{\alpha^{k/2}} \ (k = 1, 2, 3), \quad H_k = \frac{k\theta_k}{\alpha^{(k-1)/2}} \ (k = 2, 3, 4).$$

The character normalization makes their second moments independent of α. Jack coefficient orthogonality [2, Lemmas 3.1–3.2] gives

$$\mathbb{E}H_k = 0, \qquad \mathbb{E}(H_j H_k) = \mathbf{1}_{j=k}\, k n^{\downarrow k}, \qquad n^{\downarrow k} = n(n-1)\cdots(n-k+1). \tag{11}$$

In detail, the coefficient orthogonality is

$$\sum_{\lambda\vdash n}\frac{\theta^\lambda_\mu\theta^\lambda_\nu}{c_\lambda c'_\lambda}=\frac{\mathbf{1}_{\mu=\nu}}{z_\mu\alpha^{\ell(\mu)}},\qquad z_\mu=\prod_{r\ge1}r^{m_r(\mu)}m_r(\mu)!,\qquad \theta^\lambda_{(1^n)}=1.$$

For $\mu=(k,1^{n-k})$, one has $z_\mu=k(n-k)!$ and $\ell(\mu)=n-k+1$. Multiplication by $\alpha^n n!$ gives $\mathbb{E}\theta_k^2=\alpha^{k-1}n^{\downarrow k}/k$ and the stated means and cross terms.

Lassalle's low content identities [4, Section 15, p. 41 of version 3] give

$$\begin{aligned}H_2&=2P_1,\\ H_3&=3(P_2-\delta P_1-N),\\ H_4&=4[P_3-3\delta P_2+(2\delta^2-(2n-3))P_1+2\delta N].\end{aligned}\tag{12}$$

His content convention is c/α, his parameter β is $1-\alpha$, and his coefficient ϑ_k is $k\theta_k$. These substitutions give (12); only the three displayed exact identities are used. In particular $W=P_1/\sqrt{N}=H_2/(2\sqrt{N})$ has $\mathbb{E}W=0$ and $\mathbb{E}W^2=1$.

2.3 Conditional moments of the exchangeable pair

Draw $\nu\sim\pi_{n-1,\alpha}$ and, conditionally on ν, grow λ and λ' independently by the upward kernel. Define $W=s_1(\lambda)/\sqrt{\alpha N}$ and $W'=s_1(\lambda')/\sqrt{\alpha N}$. By (5) these have the required Jack marginals, and the pair is exchangeable. Conditional on λ, its construction first removes a box of content Y by the reverse kernel, and then inserts a box of content X from ν. Thus

$$\Delta:=W-W'=\frac{Y-X}{\sqrt{\alpha N}}.$$

Lemma 2.2. *For this pair,* $\mathbb{E}(\Delta\mid\lambda)=2W/n$. *Moreover*

$$A_\lambda:=1-\frac{n}{4}\mathbb{E}(\Delta^2\mid\lambda)=-\frac{H_3+\delta H_2}{4N},\tag{13}$$

$$M_\lambda:=\mathbb{E}(\Delta^3\mid\lambda)=\frac{H_4+3\delta H_3+(5n-6+b)H_2}{nN^{3/2}}.\tag{14}$$

Their second moments are

$$V:=\mathbb{E}A_\lambda^2=\frac{3(n-2)+2b}{4n(n-1)},\tag{15}$$

$$\mathbb{E}M_\lambda^2=\frac{8Q}{n^4(n-1)^2},\tag{16}$$

$$Q=4(n-2)(n-3)+27b(n-2)+2(5n-6+b)^2.\tag{17}$$

Proof. Let $D_j=\mathbb{E}(Y^j\mid\lambda)$, as in Lemma 2.1. Conditional on the deleted box, the inserted content has mean zero and variance $\alpha(n-1)$. It follows that

$$\mathbb{E}(\Delta\mid\lambda)=\frac{D_1}{\sqrt{\alpha N}}=\frac{2}{n}W,\qquad \mathbb{E}(\Delta^2\mid\lambda)=\frac{D_2+\alpha(n-1)}{\alpha N}=\frac{3P_2-\delta P_1+N}{nN}.$$

Formula (12) gives (13). For the cubic moment note that $s_1(\nu)=s_1(\lambda)-Y$. Using (6) conditional on the deleted box, we obtain

$$\begin{aligned}\mathbb{E}((Y-X)^3\mid\lambda)&=D_3+3\alpha(n-1)D_1-2\alpha(s_1-D_1)-\alpha d(n-1)\\ &=\frac{4s_3-3ds_2+d^2s_1}{n}+2\alpha s_1-\frac{\alpha d(n-1)}{2}.\end{aligned}$$

After division by $(\alpha N)^{3/2}$, the identity

$$H_4 + 3\delta H_3 + (5n - 6 + b)H_2 = 4P_3 - 3\delta P_2 + (b + 2n)P_1 - \delta N$$

proves (14). Finally, (11) gives

$$\mathbb{E}A_\lambda^2 = \frac{3n^{\downarrow 3} + 2bn^{\downarrow 2}}{16N^2}, \qquad \mathbb{E}M_\lambda^2 = \frac{4n^{\downarrow 4} + 27bn^{\downarrow 3} + 2(5n - 6 + b)^2 n^{\downarrow 2}}{n^2 N^3},$$

which reduce to (15) and (16). □

3 The uniform estimate

We use the following normal-approximation result of Shao and Zhang [5, Corollary 2.2 of version 2]. If (U, U') is a real exchangeable pair, $D = U - U'$, $\mathbb{E}(D \mid U) = \tau(U + R)$ with $0 < \tau < 1$, and $\mathbb{E}|U| \le 2$, then

$$d_{\mathrm{K}}(U, Z) \le \mathbb{E}\left|1 - \frac{1}{2\tau}\mathbb{E}(D^2 \mid U)\right| + \mathbb{E}|R| + 2\sqrt{\frac{\mathbb{E}|\mathbb{E}(D^3 \mid U)|}{\tau}}. \tag{18}$$

Our pair has finite support, so all required moments exist.

Proof of Theorem 1.1. First let $n \ge 5$. Since W is a function of λ, Lemma 2.2 gives $\mathbb{E}(\Delta \mid W) = (2/n)W$. Also $\mathbb{E}|W| \le 1$, because $\mathbb{E}W^2 = 1$. We may therefore use (18) with $\tau = 2/n$ and $R = 0$. Conditioning on λ gives the defect A_λ and cubic moment M_λ in Lemma 2.2. Conditional expectation and Cauchy–Schwarz imply

$$\mathbb{E}\left|1 - \frac{n}{4}\mathbb{E}(\Delta^2 \mid W)\right| \le \sqrt{V}, \qquad \mathbb{E}|\mathbb{E}(\Delta^3 \mid W)| \le \mathbb{E}|M_\lambda| \le \|M_\lambda\|_2.$$

Consequently

$$d_{\mathrm{K}}(W, Z) \le \sqrt{V} + 2\sqrt{\|M_\lambda\|_2/(2/n)}. \tag{19}$$

Put $t = \alpha/n$ and $u = b/n$. Since $b = (\alpha - 1)^2/\alpha \le \alpha$ for $\alpha \ge 1$, one has $0 \le u \le t$. The exact expressions in (15)–(17) give

$$V \le \frac{3 + 2u}{4(n - 1)} \le \frac{5\max\{1, t\}}{4(n - 1)},$$
$$Q \le 54n^2 + 47bn + 2b^2 = n^2(54 + 47u + 2u^2) \le 103n^2 \max\{1, t^2\}.$$

For the last inequality, separate $u \le 1$ and $u \ge 1$ and bound each term by its coefficient times $\max\{1, u^2\}$. Substituting into (16) and (19) yields

$$d_{\mathrm{K}}(W, Z) \le \left(\frac{\sqrt{5}}{2} + 2\,206^{1/4}\right)\frac{\max\{1, \sqrt{t}\}}{\sqrt{n - 1}}.$$

Let $r = \max\{n^{-1/2}, \sqrt{\alpha}/n\} = \max\{1, \sqrt{t}\}/\sqrt{n}$. For $n \ge 5$, $\sqrt{n/(n-1)} \le \sqrt{5/4} < 9/8$. Since $\sqrt{5}/2 < 9/8$ and $206^{1/4} < 19/5$, the preceding bound is less than

$$\frac{9}{8}\left(\frac{9}{8} + \frac{38}{5}\right)r = \frac{3141}{320}r < 10r.$$

For $n = 3, 4$, use $d_{\mathrm{K}}(W, Z) \le 1$ and $r \ge n^{-1/2} \ge 1/2$. This proves (2) for all the stated parameters. □

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